

Credit Risk Classification and Model Evaluation

Summary

This report presents an end-to-end machine learning analysis aimed at predicting credit risk. Using a dataset of 1,000 customer records, two classification models: Logistic Regression and a Decision Tree, were trained to identify 'Good' versus 'Bad' loan applicants. Both models achieved an identical overall accuracy of 72.67%. However, a detailed analysis of the confusion matrices reveals that the Decision Tree model is slightly more conservative, making it the recommended choice for a financial institution seeking to minimize the costly risk of approving loans to likely defaulters.

Dataset Overview

The dataset contains 1,000 records with 21 distinct financial and demographic attributes (e.g., Account Balance, Duration of Credit, Purpose, Age).

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- Data Quality:
 - Completeness: The data is highly pristine with zero missing (null) values across all 21 columns, eliminating the need for imputation.
 - Target Variable: The target variable is 'Creditability', where 1 indicates a 'Good Risk' (credible) and 0 indicates a 'Bad Risk' (default prone).

Exploratory Data Analysis (EDA)

Before modeling, exploratory analysis was conducted to understand the underlying risk profiles:

Completeness: The data is highly pristine with zero missing (null) values across all 21 columns, eliminating the need for imputation.

Target Variable: The target variable is 'Creditability', where 1 indicates a 'Good Risk' (credible) and 0 indicates a 'Bad Risk' (default prone).

Class Imbalance: A count plot of the target variable revealed an imbalance typical in risk detection. Approximately 700 applicants are Good Risks (1), while 300 are Bad Risks (0).

Modeling Methodology

- **Feature Selection:** For this baseline modeling phase, three core continuous features were selected: 'Age_years', 'Credit_Amount', and 'Duration_of_Credit_monthly'.
- **Data Partitioning:** The data was split using a 70/30 train-test split, reserving 30% of the data (300 records) to evaluate the models on unseen data.
- **Algorithms Used:**
 - **Logistic Regression:** Used as a baseline weighted scorecard.
 - **Decision Tree Classifier:** Configured with a max_depth of 3 to prevent overfitting and maintain interpretability.

Model Evaluation & Results

Both models were evaluated on the 30% holdout test set. While both achieved an accuracy of 72.67%, their underlying performance varies significantly when analyzing the Confusion Matrix.

Logistic Regression Performance

Accuracy: 72.67%

Confusion Matrix	Predicted Bad (0)	Predicted Good (1)
Actual Bad (0)	12 (True Negative)	70 (False Positive)
Actual Good (1)	12 (False Negative)	206 (True Positive)

Decision Tree Performance

Accuracy: 72.67%

Confusion Matrix	Predicted Bad (0)	Predicted Good (1)
Actual Bad (0)	15 (True Negative)	67 (False Positive)
Actual Good (1)	15 (False Negative)	203 (True Positive)

Business Recommendation

Although both models successfully predicted 218 out of 300 test cases correctly, the types of errors they made differ considerably.

The Decision Tree is the superior model for this specific business context. In banking, a 'False Positive' (approving a bad loan) is significantly more expensive than a 'False Negative' (rejecting a good loan). The Decision Tree successfully rejected 3 more bad loans (True Negatives increased from 12 to 15) and reduced dangerous False Positives from 70 to 67 compared to the Logistic Regression model.